14.2 answers

1. As long as $f$ is <u>continuous</u>, $f(2,3) = 8$

2. a) 679

   b) 3

   c) DNE

   d) 0

   e) DNE

   f) 1298

3. $\{(x,y) \mid y \geq -\frac{2}{3}x + 2\}$

4. a) $\{(x,y) \mid y > -x + 2\}$

   Above the line $y = -x + 2$

   b) $\{(x,y) \mid x^2 + y^2 \leq 1\}$

   Inside and on the circle $x^2 + y^2 = 1$

   c) $\{(x,y) \mid x^2 + y^2 > 4\}$

   Outside the circle $x^2 + y^2 = 4$

   d) $f$ is <u>not</u> continuous at $(0,0)$, so the largest set on which $f$ is continuous is $\{(x,y) \mid (x,y) \neq (0,0)\}$

5. 0

6a) $0$

b) since $\lim_{(x,y)\to(0,0)} g(x,y) = 0 = f(0,0)$, $g$ is contiuous at $(0,0)$.